

Nianhua(Mason) Yuan

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Education

University of Illinois Urbana-Champaign, Grainger College of Engineering Sep 2024 – Dec 2025
Master of Science in Financial Engineering

- **Related Coursework:** Financial Computing, Machine Learning, Deep Learning, Reinforcement Learning, Stochastic Calculus.

Rensselaer Polytechnic Institute, College of Mathematical Sciences Jan 2021 – May 2024
Bachelor of Science in Mathematics

- **Dual Major:** Applied Mathematics, Economics.
- **Related Coursework:** Econometric Methods for Big Data, Math Models of Operations Research, Numerical Computing, etc.

Skills

- **Programming Language:** Python, R, SQL, C++, AMPL
- **Frameworks:** Pandas, NumPy, Jax, Matplotlib, Torch, QuantLib, Scikit-Learn
- **Certificates:** FRM Level 1 Certificate, CFA Level 1 Certificate

Experience

China Futures Co. — Quantitative Research Intern (Beijing, China) May 2025 – Aug 2025
Worked with the research team on data analysis and model evaluation for futures and factor-based trading strategies.

- Processed and organized daily futures market data to support strategy backtesting.
- Evaluated factor effectiveness using historical data and contributed to regular performance reviews.
- Assisted in developing a simplified machine-learning workflow for portfolio weight adjustment under various market conditions.
- Communicated findings through concise reports and data visualizations used by senior analysts.
- **Skills:** Python · Data Analysis · Time-Series Evaluation · Financial Reporting

J.P. Morgan Practicum — Financial Data Analyst (Urbana, IL) Jan 2025 – May 2025
Collaborated with a small team to improve options data quality and volatility-surface calibration for SPX/VIX.

- Cleaned and integrated large SPX/VIX option-chain datasets; standardized fields and built checks for outliers and missing quotes.
- Estimated forward prices using spot, rates, and dividends; applied interpolation and imputation for sparse strikes and maturities.
- Implemented calibration scripts (weighted least squares) to fit volatility surfaces and compared fit quality across trading days.
- Improved model stability by weighting tighter bid-ask contracts and filtering illiquid short-dated options.
- Summarized results with charts and diagnostics for weekly reviews; documented assumptions, data issues, and next steps.
- **Skills:** Python · Data Cleaning · Options Data · Financial Modeling · Reporting & Visualization

Bosera Asset Management Co., Ltd. – Financial Data & Client Reporting Intern (Hongkong) May 2023 – Aug 2023
Supported the retail investment division by assisting in data tracking, fund performance reporting, and client communication.

- Maintained and updated daily fund performance reports using Excel and the Wind plugin, ensuring data accuracy across multiple funds.
- Conducted return attribution and peer comparison analysis to identify performance drivers and market positioning.
- Prepared client updates summarizing fund performance, portfolio allocation, and market commentary.
- Assisted in investor engagement events such as "Meet the Fund Manager" and "Bosera Open Day", drafting meeting notes and summarizing client feedback for internal review.
- **Skills:** Excel · Financial Analysis · Reporting & Documentation · Client Support · Market Research

Academic Projects

Machine Learning Application for Option Pricing Sep 2024 – Dec 2024

- **Data Generation:** Generated sample dataset using pricing models to simulate option prices under various market conditions.
- **Model Selection and Optimization:** Used the dataset to train **machine learning** models, including Random Forest, LightGBM, and Deep Neural Networks (DNNs). Employed grid search to optimize hyperparameters.